

MD RIYAU HAQUE

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Professional Summary

M.Tech researcher in **Fuel & Energy Engineering** at IIT (ISM) Dhanbad with **8+ years of mechanical engineering experience** across India and the GCC, covering refinery QA/QC, mechanical inspection, NDT, equipment integrity, HVAC/MEP, maintenance, and technical project coordination. Experienced in **stage-wise inspection, refinery equipment assessment, NDT methods, shutdown activities, Management of Change (MOC), Hot Tapping, WOLE, hydrotesting, and HSE compliance**. ASNT Level-II certified in **VT, DPT/PT, UT, RT, and MPI/MT**, with GATE 2025 qualification (All India Score: 418). Current technical focus includes **hydrogen production, hydrogen reforming, biomass conversion, pyrolysis, thermochemical conversion, process modeling, mass and energy balances, and energy systems**, with working knowledge of **Aspen Plus, MATLAB, CFD, AutoCAD, and MS Excel**. Targeting roles in **Energy Systems, Hydrogen/Clean Energy, Process Engineering, Mechanical Inspection, Refinery QA/QC, Equipment Integrity, and Maintenance Engineering**.

Education

Indian Institute of Technology (ISM), Dhanbad

M.Tech in Fuel & Energy Engineering – CGPA: 7.11/10

YBN University, Ranchi

B.Tech in Mechanical Engineering – CGPA: 7.41/10

G B Pant Government of Delhi Polytechnic

Diploma in Mechanical Engineering – 71.34%

Dhanbad, Jharkhand

Expected May 2027

Ranchi, Jharkhand

Completed 2025

Delhi, India

Completed 2014

Relevant Coursework & Technical Training

Energy Systems, Hydrogen Production & Reforming, Biomass Pyrolysis, Biochar Production, Thermochemical Conversion, Equipment Integrity Assessment, Mechanical Inspection, NDT Methodology, HSE Compliance, Process Modeling & Simulation, CFD Analysis, MATLAB Programming, Mass & Energy Balances, Combustion Engineering, Renewable Energy Systems

Experience

Mechanical QA/QC Engineer – Shutdown

May 2026 – Jun 2026

Reliance Industries, Jamnagar, Gujarat – CDU Unit, through CR3 India Ltd

Jamnagar, Gujarat

- Executed **mechanical QA/QC inspection, maintenance, and equipment integrity** activities throughout a **45-day refinery shutdown**.
- Performed **7 inspection and maintenance methods**: Visual Testing (VT), Dye Penetrant Testing (DPT), Magnetic Particle Inspection (MPI), Radiographic Testing (RT), Eddy Current Testing (ECT), hydrotesting, and hydro jetting, with PWHT inspection where applicable.
- Conducted **stage-wise mechanical inspection** across **6 major equipment categories**: heat exchangers, pressure vessels, air-fin-fan coolers, columns, fired heaters, and storage tanks.
- Inspected equipment and reviewed execution requirements associated with **Hot Tapping, WOLE, and Management of Change (MOC)** activities.
- Oversaw maintenance and MOC-related activities across **8 refinery operating areas**, including Crude-2, Sulphur, LDPE, PP, CPF, MTF, SWRO, and Jetty Oil Terminal.

Project Coordinator – MEP

Dec 2019 – May 2022

Buraimi & Sohar Regional Sports Complexes – through Genetco

Oman

- Directed **preventive and breakdown maintenance** across **3 MEP workstreams**: mechanical, electrical, and civil.

- Supervised maintenance activities for **3 building-service systems**: HVAC, plumbing, and fire-fighting, including site inspection and maintenance coordination.
- Managed contractors, vendors, manpower allocation, material requirements, and work schedules to support project execution and facility operations.
- Maintained **HSE compliance** through routine site inspections, safety coordination, and execution-team communication.
- Prepared **MIS and progress reports** while tracking project budgets, operational costs, work progress, and execution status.

General Foreman – HVAC Division

Aug 2018 – Jul 2019

EXPO 2020 Project – Al-Futtaim Engineering

Dubai, UAE

- Coordinated installation, testing, commissioning, and troubleshooting of **central air-conditioning systems** for the EXPO 2020 project.
- Supervised foremen and technicians while coordinating materials, equipment, manpower, and site execution activities.
- Monitored **preventive maintenance schedules, HVAC quality requirements, and installation activities** against client specifications.
- Prepared daily progress and pre-commissioning reports supporting **project tracking, testing, commissioning, and system handover**.

HVAC Supervisor

Oct 2016 – Apr 2018

Saudi Arabia Army Camp – Ministry of Defence Projects, through Beijing Construction Engineering International

Saudi Arabia

- Inspected, maintained, and commissioned **centralized HVAC systems** across Saudi Arabia Army Camp facilities.
- Diagnosed HVAC failures and performed **breakdown troubleshooting** to restore system operation across assigned facilities.
- Managed HVAC inventory and spare-parts planning to improve maintenance responsiveness and equipment availability.
- Maintained compliance with applicable **safety, quality, and site requirements** during maintenance and troubleshooting activities.
- Investigated HVAC failures, identified technical issues, and recommended **repair or replacement solutions** based on equipment condition and service requirements.

Service Engineer

2014 – 2016

Automotive Service Operations

India

- Managed technical service requirements and **customer technical escalations** across multiple states.
- Supported **AMC target achievement** and coordinated dealer and franchise service operations.
- Delivered technical training to service partners and resolved escalation cases in line with defined **SLA and KPI requirements**.

Projects

Comparative Experimental Investigation of Rice Husk and Sawdust Pyrolysis | *Biomass Pyrolysis, Biochar, Process Optimization*

- Conducting a comparative experimental investigation of **rice husk and sawdust pyrolysis** for biochar production and renewable energy applications.
- Evaluating the effects of **pyrolysis temperature, heating rate, and residence time** on biochar yield and fuel properties.
- Characterizing biomass feedstocks and biochar using **moisture, volatile matter, ash, fixed carbon, higher heating value (HHV), and bulk density**.
- Applying **mass-balance-based yield analysis** and comparative performance assessment to evaluate operating conditions and feedstock suitability.
- Assessing pyrolysis for **waste valorization, renewable energy, circular economy, and sustainable biochar applications**.

Certifications

ASNT Level-II NDT Certification: Visual Testing (VT), Liquid Penetrant Testing (DPT/PT), Ultrasonic Testing (UT), Radiographic Testing (RT), Magnetic Particle Testing (MPI/MT).

QA/QC Mechanical Inspector: Mechanical QA/QC inspection, quality-control activities, equipment inspection, and inspection practices.

Certificate of Facility Manager: Indian Institute of Skill Development Training, Bengaluru.

Technical Skills

Energy & Process Engineering: Energy Systems, Hydrogen Production, Hydrogen Reforming, Clean Energy, Decarbonization, Process Engineering, Process Modeling, Thermochemical Conversion, Biomass Conversion, Biomass Pyrolysis, Biochar Production, Energy Efficiency, Renewable Energy, Mass & Energy Balances, Combustion Engineering
Mechanical QA/QC, Inspection & NDT: Mechanical QA/QC, Mechanical Inspection, Equipment Integrity, Stage-Wise Inspection, ASNT Level-II, Visual Testing (VT), Liquid Penetrant Testing (DPT/PT), Ultrasonic Testing (UT), Radiographic Testing (RT), Magnetic Particle Testing (MPI/MT), Eddy Current Testing (ECT), PWHT, Hydrotesting, Hydro Jetting
Refinery Equipment & Shutdown: Heat Exchangers, Pressure Vessels, Columns, Fired Heaters, Air-Fin-Fan Coolers, Storage Tanks, CDU Equipment, Refinery Shutdown, Hot Tapping, WOLE, Management of Change (MOC), Equipment Integrity Assessment
HVAC & MEP Engineering: Chillers, VRF Systems, Package Units, Air Handling Units (AHU), Fan Coil Units (FCU), Central Air Conditioning, HVAC Maintenance, Plumbing, Fire-Fighting Systems, Testing, Commissioning, Troubleshooting
Software & Engineering Tools: Aspen Plus, MATLAB, CFD, AutoCAD, MS Excel, MS Office, C Programming
Operations, Maintenance & Management: Preventive Maintenance, Breakdown Maintenance, Contractor Management, Vendor Management, Manpower Planning, Maintenance Coordination, MIS Reporting, SLA/KPI Management, AMC Operations, HSE Coordination, Technical Reporting

Achievements

GATE 2025 – Mechanical Engineering: Qualified with an **All India Score of 418**.
Inter IIT Weightlifting Championship 2025: Secured **6th Position**.
District Weightlifting Championship 2025: Won **Silver Medal** in Dhanbad.
Inter IIT Sports – March Past 2025: Won **Bronze Medal**.
Inter Delhi Polytechnic Kabaddi Tournament 2012: Won **Silver Medal**.

Leadership

Class Representative & Placement Cell Coordinator – B.Tech & Diploma: Coordinated academic communication, student coordination, and placement-related activities; facilitated recruitment opportunities for **95+ students** across multiple companies during the respective academic programs.

Languages

Languages: English, Hindi, Urdu, Arabic